Reti Antonio

30232

Mini-StackOverflow
Supplementary Specification
Version 1.0

Revision History

Date	Version	Description	Author
30/03/2026	1.0	First Version	Reti Antonio

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1. Introduction

The purpose of this document is to specify the non-functional requirements and design constraints for the Mini-StackOverflow project. This system is a client-server application designed to support two distinct user roles: Regular Users and Moderators. This document ensures the system meets professional standards for performance, security, and reliability.

2. Non-Functional Requirements

2.1 Performance Requirements

- **Search Latency**: The system must retrieve question search results from the database within 2 seconds for a standard query.
- **Concurrency**: The server must support at least 50 concurrent users performing actions like voting or posting without a degradation in response time.

2.2 Security Requirements

- **Role-Based Access Control (RBAC)**: The system must strictly enforce permissions so that only Moderators can access administrative functions such as user suspension and post deletion.
- **Authentication**: Users must be authenticated before they are permitted to create questions, post answers, or edit their own posts.
- **Data Integrity**: Only the original author of a post is permitted to edit their content; the system must prevent unauthorized modification of peer posts.
- **Vulnerability Mitigation:** The application must be developed following industry-standard security practices to mitigate **Common Vulnerabilities and Exposures (CVEs)**.

2.3 Usability Requirements

- **Interface Clarity**: The search interface must be prominent and easy to use to encourage users to find existing answers before creating new questions.
- **Error Handling**: Users should receive clear feedback if an action fails, such as attempting to post an answer while suspended.
- **Content Moderation Workflow**: The Moderator dashboard must provide a clear view of flagged posts to facilitate quick approval or deletion.

2.4 Reliability & Availability

- **Data Persistence**: The system must store all data in a persistent database to ensure no information is lost in the event of a server restart.
- **Uptime**: The application should maintain 99% availability during business hours to support active community discussions.

2.5 Maintainability

- **Modularity**: The system design should allow for the addition of new tags or categories without requiring changes to the core server logic.